

```
import pandas as pd
```

```
df=pd.read_csv("/content/diabetes (1).csv")
```

✓ EXPLORE THE DATASET

```
# to see top 5 rows from starting  
df.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabetes
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
# to see particular rows from starting  
df.head(3)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabetes
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.tail()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabet
763	10	101	76	48	180	32.9	
764	2	122	70	27	0	36.8	
765	5	121	72	23	112	26.2	
766	1	126	60	0	0	30.1	
767	1	93	70	31	0	30.4	

```
# to see the shape of the dataset
df.shape
```

```
(768,9)
```

```
(768, 9)
```

```
# to see the name of the coloumn
df.columns
```

```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness',
       'Insulin',
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='object')
```

```
# Datatypes of the column
df.dtypes
```

	0
Pregnancies	int64
Glucose	int64
BloodPressure	int64
SkinThickness	int64
Insulin	int64
BMI	float64
DiabetesPedigreeFunction	float64
Age	int64
Outcome	int64

```
dtype: object
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
```

```
Data columns (total 9 columns):
```

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	int64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	int64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

```
dtypes: float64(2), int64(7)
```

```
memory usage: 54.1 KB
```

```
df.describe()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.999999
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.999999
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.500000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.000000

✓ CHECK DATA QUALITY

```
df.isnull()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabe
0	False	False	False	False	False	False	
1	False	False	False	False	False	False	
2	False	False	False	False	False	False	
3	False	False	False	False	False	False	
4	False	False	False	False	False	False	
...	...	...	...	...	...	...	
763	False	False	False	False	False	False	
764	False	False	False	False	False	False	
765	False	False	False	False	False	False	
766	False	False	False	False	False	False	
767	False	False	False	False	False	False	

768 rows × 9 columns

```
#check the duplicate value
df.duplicated().sum()
```

```
np.int64(0)
```

```
# to count the unique values
df.nunique()
```

	0
Pregnancies	17
Glucose	136
BloodPressure	47
SkinThickness	51
Insulin	186
BMI	248
DiabetesPedigreeFunction	517
Age	52
Outcome	2

dtype: int64

```
# to check the invalid zero values(real missing value)
(df[['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']] == 0).sum()
```

	0
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11

dtype: int64

```
# select a column  
df ['Glucose']
```

	Glucose
0	148
1	85
2	183
3	89
4	137
...	...
763	101
764	122
765	121
766	126
767	93

768 rows × 1 columns

dtype: int64

```
#multiple coloumn  
df[['Age', 'BloodPressure']]
```

	Age	BloodPressure
0	50	72
1	31	66
2	32	64
3	21	66
4	33	40
...	...	...
763	63	76
764	27	70
765	30	72
766	47	60
767	23	70

768 rows × 2 columns

```
#filter row by some condition  
df[df['Age']>60].count()
```

	0
Pregnancies	27
Glucose	27
BloodPressure	27
SkinThickness	27
Insulin	27
BMI	27
DiabetesPedigreeFunction	27
Age	27
Outcome	27

dtype: int64

```
# filter with multiple condition  
df[(df['Age']>60) & (df['BMI']>30)]
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabet
115	4	146	92	0	0	31.2	
148	5	147	78	0	0	33.7	
221	2	158	90	0	0	31.6	
263	3	142	80	15	0	32.4	
362	5	103	108	37	0	39.2	
363	4	146	78	0	0	38.5	
579	2	197	70	99	0	34.7	
666	4	145	82	18	0	32.5	
674	8	91	82	0	0	35.6	
759	6	190	92	0	0	35.5	
763	10	101	76	48	180	32.9	

```
df[(df['Age']>65) & (df['Outcome']==1)]
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabet
221	2	158	90	0	0	31.6	
363	4	146	78	0	0	38.5	
666	4	145	82	18	0	32.5	
759	6	190	92	0	0	35.5	

```
# access rows by index
df.iloc[2:5]
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabetes
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

```
#Sort by using a column
df.sort_values(by='Age')
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabet
736	0	126	86	27	120	27.4	
738	2	99	60	17	160	36.6	
721	1	114	66	36	200	38.1	
60	2	84	0	0	0	0.0	
290	0	78	88	29	40	36.9	
...	...	...	...	...	...	...	
123	5	132	80	0	0	26.8	
684	5	136	82	0	0	0.0	
666	4	145	82	18	0	32.5	
453	2	119	0	0	0	19.6	
459	9	134	74	33	60	25.9	

768 rows × 9 columns

df['Glucose'].value_counts()

	count
Glucose	
99	17
100	17
111	14
125	14
129	14
...	...
56	1
169	1
149	1
65	1
190	1

136 rows × 1 columns

dtype: int64

```
# group by and aggregate the data
df.groupby('Outcome').agg({'Glucose': 'mean', 'BMI': 'median'})
```

	Glucose	BMI
Outcome		
0	109.980000	30.05
1	141.257463	34.25

```
df.groupby('Outcome').mean()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	
Outcome						
0	3.298000	109.980000	68.184000	19.664000	68.792000	30
1	4.865672	141.257463	70.824627	22.164179	100.335821	35

```
#Statistics  
# Mean of the data  
df['BMI'].mean()
```

```
np.float64(31.992578124999998)
```

```
#median  
df['BMI'].median()
```

```
32.0
```

```
#Standard Deviation  
df['Age'].std()
```

```
11.76023154067868
```

```
# correlation matrix  
df.corr()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness
Pregnancies	1.000000	0.129459	0.141282	-0.081672
Glucose	0.129459	1.000000	0.152590	0.057328
BloodPressure	0.141282	0.152590	1.000000	0.207371
SkinThickness	-0.081672	0.057328	0.207371	1.000000
Insulin	-0.073535	0.331357	0.088933	0.436783
BMI	0.017683	0.221071	0.281805	0.392573
DiabetesPedigreeFunction	-0.033523	0.137337	0.041265	0.183928
Age	0.544341	0.263514	0.239528	-0.113970
Outcome	0.221898	0.466581	0.065068	0.074752

```
# add new coloumn
df['BMI_CATEGORY']=pd.cut(df['BMI'],bins=[0,18.5,24.9,29.9,100],labels=['Und
```

```
df.head(20)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabetes
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
Next steps:	Generate code with df		New interactive sheet				